$$y_{(x=x_0+nh)} = y_0 + \frac{n}{1!}\Delta y_0 + \frac{n(n-1)}{2!}\Delta^2 y_0 + \frac{n(n-1)(n-2)}{3!}\Delta^3 y_0 + \ldots$$

To find y at $x = 45$ $\therefore x_0 + nh = 45$, $x_0 = 40, h = 10$ $\Rightarrow n = \frac{1}{2}$

$$y_{(x=45)} = 31 + \frac{1}{2} \times 42 + \frac{\frac{1}{2}\left(\frac{-1}{2}\right)}{2}(9) + \frac{\frac{1}{2}\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)}{6} \times (-25) + \frac{\frac{1}{2}\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)\left(\frac{-5}{2}\right)}{24} \times (37)$$

$$= 31 + 21 - \frac{9}{8} - \frac{25}{16} - \frac{37 \times 15}{384}$$

$$= 47.867 \cong 48$$

Example 5.16

Using appropriate interpolation formula find the number of students whose weight is between 60 and 70 from the data given below:

Weight in lbs	0-40	40-60	60-80	80-100	100-120
No. of Students	250	120	100	70	50

Solution:

Let x be the weight and y be the number of students.

Difference table of cumulative frequencies are given below.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
Below 40	250				
		120			
60	370		–20		
		100		–10	
80	470		–30		20
		70		10	
100	540		–20		
		50			
120	590				

Let us calculate the number of students whose weight is below 70. For this we use forward difference formula.

$$y_{(x=x_0+nh)} = y_0 + \frac{n}{1!}\Delta y_0 + \frac{n(n-1)}{2!}\Delta^2 y_0 + \frac{n(n-1)(n-2)}{3!}\Delta^3 y_0 + \ldots$$

To find y at $x = 70$ $\therefore$ $x_0 + nh = 70$, $x_0 = 40, h = 20$

$$40 + n(20) = 70 \quad \Rightarrow n = 1.5$$

$$\therefore \; y_{(x=70)} = 250 + 1.5(120) + \frac{(1.5)(0.5)}{2!}(-20) + \frac{(1.5)(0.5)(-0.5)}{3!}(-10) + \frac{(1.5)(0.5)(-0.5)(-1.5)}{4!}(20)$$

$$= 250 + 180 - 7.5 + 0.625 + 0.46875$$

$$= 423.59$$

$$\cong 424$$

Number of students whose weight is between 60 and $70 = y(70) - y(60) = 424 - 370 = 54$

Example 5.17

The population of a certain town is as follows:

Year : x	1941	1951	1961	1971	1981	1991
Population in lakhs : y	20	24	29	36	46	51

Using appropriate interpolation formula, estimate the population during the period 1946.

Solution:

x	1941	1951	1961	1971	1981	1991
y	20	24	29	36	46	51

Here we find the population for year1946. (i.e) the value of y at x=1946. Since the value of y is required near the beginning of the table, we use the Newton's forward interpolation formula.

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